

Sec 7. Path Integral for Harmonic Oscillator ①

Consider a harmonic oscillator with Hamiltonian

$$H(P, Q) = \frac{1}{2m} P^2 + \frac{1}{2} m \omega^2 Q^2$$

$$\langle 0|0 \rangle_f = \int DQ DP Dq e^{-i \int_{-\infty}^{+\infty} dt [p \dot{q} - (1-i\epsilon)H + f q]}$$

the $1-i\epsilon$ can be traded as

$$\frac{1}{2m} \rightarrow \frac{1}{2m(1+i\epsilon)}, \quad \frac{1}{2} m \rightarrow \frac{1}{2} (1-i\epsilon) m$$

$$\dot{q} = \frac{\partial H}{\partial p}$$

using the Lagrangian heartment:

$$\langle 0|0 \rangle_f = \int DQ Dq e^{-i \int_{-\infty}^{+\infty} dt \left[\frac{1}{2} (1+i\epsilon) m \dot{q}^2 - \frac{1}{2} (1-i\epsilon) m \omega^2 q^2 + f q \right]}$$

from now on, we simply set $m=1$, this can be restored by dimensional analysis.

1 The p integral is Gaussian, stationary at $p = m\dot{q}/(1-i\epsilon)$, giving $\frac{m\dot{q}^2}{2(1-i\epsilon)} \simeq \frac{1}{2}(1+i\epsilon)m\dot{q}^2$. Substituting $p = m\dot{q}$ gives the same to $O(\epsilon)$, since the exponent is stationary in p . The q -independent prefactor goes into Dq .

Let use Fourier transformed variables

$$\tilde{q}(E) = \int_{-\infty}^{+\infty} dt e^{iEt} q(t), \quad q(t) = \int_{-\infty}^{+\infty} \frac{dE}{2\pi} e^{-iEt} \tilde{q}(E)$$

The exponential part then becomes

$$\int_{-\infty}^{+\infty} dt \frac{1}{2} \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \frac{dE'}{2\pi} e^{-i(E+E')t} [-(1+i\epsilon)EE' - (1-i\epsilon)\omega^2] \tilde{q}(E) \tilde{q}(E') + \tilde{f}(E) \tilde{q}(E') + \tilde{f}(E') \tilde{q}(E)$$

Integrating over dt

(2)

$$S = \int_{-\infty}^{+\infty} dt \dots \xrightarrow{\text{redefine } E} \frac{1}{2} \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \left[\overbrace{(1+i\epsilon)E^2 - (1-i\epsilon)\omega^2}^{E^2 - \omega^2 + i(E^2 + \omega^2)\epsilon} \tilde{q}(E) \tilde{q}(-E) + \tilde{f}(E) \tilde{q}(E) + \tilde{f}(-E) \tilde{q}(-E) \right] \quad \text{free theory}$$

$$= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \left[(E^2 - \omega^2 + i\epsilon) \tilde{q}(E) \tilde{q}(-E) + \tilde{f}(E) \tilde{q}(-E) + \tilde{f}(-E) \tilde{q}(E) \right]$$

"symmetrize" the variables.

$$\tilde{\chi}(E) = \tilde{q}(E) + \frac{f(E)}{E^2 - \omega^2 + i\epsilon} \quad Dq \rightarrow Dx.$$

$$S = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \left[\tilde{\chi}(E) (E^2 - \omega^2 + i\epsilon) \tilde{\chi}(-E) - \frac{\tilde{f}(E) \tilde{f}(-E)}{E^2 - \omega^2 + i\epsilon} \right]$$

As $\frac{\tilde{f}(E)}{E^2 - \omega^2 + i\epsilon}$ is only a constant ^{function.} shift.

$$Dq = Dx$$

$$\langle 0|0 \rangle_f = \exp \left[\frac{i}{2} \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \frac{\tilde{f}(E) \tilde{f}(-E)}{-E^2 + \omega^2 - i\epsilon} \right] \times$$

$$\left[Dx \exp \left[\frac{i}{2} \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \tilde{\chi}(E) (E^2 - \omega^2 + i\epsilon) \tilde{\chi}(-E) \right] \right]$$

already there for free theory

and free theory has $\langle 0|0 \rangle_{f=0} = 1$ ¹

$$\langle 0|0 \rangle_f = \exp \left[\frac{i}{2} \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \frac{\tilde{f}(E) \tilde{f}(-E)}{-E^2 + \omega^2 - i\epsilon} \right]$$

1 The shift is q -independent, so $Dx = Dq$; it is complex, but the $i\epsilon$ damping lets the contour move back. What remains, $\propto [\det(-\partial_t^2 - \omega^2 + i\epsilon)]^{-1/2}$, is f -independent, and $\langle 0|0 \rangle_{f=0} = 1$ fixes it.

rewrite in time-domain variables

$$\langle 0|0 \rangle_f = \exp \left[\frac{i}{2} \int_{-\infty}^{+\infty} dt dt' f(t) G(t-t') f(t') \right]$$

with

$$G(t-t') = \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \frac{e^{-iE(t-t')}}{-E^2 + \omega^2 - i\epsilon}$$

2 Poles at $E = \pm(\omega - i\epsilon')$: $+\omega$ just below the axis, $-\omega$ just above. For $t > t'$ close below (picks $+\omega$), for $t < t'$ above. The equation of motion fixes G only up to $e^{\pm i\omega t}$ terms; the $i\epsilon$ selects this time-ordered (Feynman) solution.

as $f(t) = \int dE e^{-iEt} \tilde{f}(E)$ ①

1 Missing $1/2\pi$: $f(t) = \int \frac{dE}{2\pi} e^{-iEt} \tilde{f}(E)$ (p. 1), and likewise for $f(t')$. With it restored the result is as written.

$$f(t') = \int dE e^{-iEt'} \tilde{f}(E)$$

Now $G(t-t')$ is a Green's function for the oscillator equation of motion:

$$\left(\frac{\partial^2}{\partial t^2} + \omega^2 \right) G(t-t') = \delta(t-t')$$

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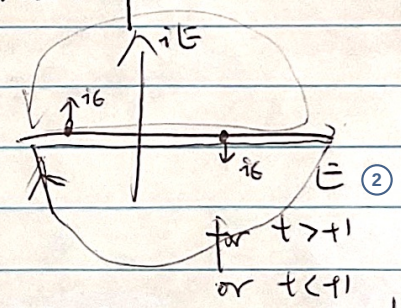
as

$$\begin{aligned} \left(\frac{\partial^2}{\partial t^2} + \omega^2 \right) G(t-t') &= \int_{-\infty}^{+\infty} \frac{dE}{2\pi} \frac{(-E^2 + \omega^2) e^{-iE(t-t')}}{-E^2 + \omega^2 - i\epsilon} \\ &= \delta(t-t') \end{aligned}$$

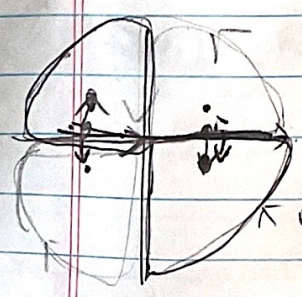
note $\delta(t-t') = \delta(t'-t)$

one can also find that, through ~~residue~~ residue theorem $= \delta(t-t')$

$$G(t-t') = \frac{i}{2\omega} \exp(-i\omega|t-t'|)$$



for $t > t'$
or $t < t'$



ensure vanishing ($t > t'$)

note it is symmetric function of t and t' (you can also see time ordering comes in here).

Recall

$$\langle 0 | T Q(t_1) \dots | 0 \rangle = \frac{1}{i} \frac{\delta}{\delta f(t_1)} \dots \langle 0 | 0 \rangle_f \Big|_{f=0}$$

with our new formula. here

¹ f appears twice in $\frac{i}{2} \int f G f$,
so $\frac{\delta}{\delta f(t_2)}$ gives $\frac{i}{2} \int dt' [G(t_2 - t') + G(t' - t_2)] f(t') =$
 $i \int dt' G(t_2 - t') f(t')$,
because G is even (p. 3).

$$\langle 0 | T Q(t_1) Q(t_2) | 0 \rangle = \frac{1}{i} \frac{\delta}{\delta f(t_1)} \frac{1}{i} \frac{\delta}{\delta f(t_2)} \langle 0 | 0 \rangle_f \Big|_{f=0}$$

$$= \frac{1}{i} \frac{\delta}{\delta f(t_1)} \left[\int_{-\infty}^{+\infty} dt' G(t_2 - t') f(t') \right] \langle 0 | 0 \rangle_f \Big|_{f=0} \quad (1)$$

$$= \left[\frac{1}{i} G(t_2 - t_1) + \int_{-\infty}^{+\infty} dt' G(t_2 - t') f(t') \int_{-\infty}^{+\infty} dt'' G(t' - t_1) f(t'') \right] \langle 0 | 0 \rangle_f \Big|_{f=0}$$

$$= \frac{1}{i} G(t_2 - t_1).$$

So it is clear we need even numbers of Q so the transition amplitude is not zero as $f=0$.

$$\langle 0 | T Q(t_1) Q(t_2) Q(t_3) Q(t_4) | 0 \rangle = \frac{1}{i^2} \left[G(t_1 - t_2) G(t_3 - t_4) + G(t_1 - t_3) G(t_2 - t_4) + G(t_1 - t_4) G(t_2 - t_3) \right]$$

and.

$$\langle 0 | T Q(t_1) \dots Q(t_{2n}) | 0 \rangle = \frac{1}{i^n} \sum_{\text{all pairings}} G(t_{i_1} - t_{i_2}) \dots G(t_{i_{n-1}} - t_{i_{2n}}) \quad (2)$$

$i_1, i_2, \dots, i_{2n}$

² At $f=0$, $2n$ derivatives see only $\frac{1}{n!} \left(\frac{i}{2} \int f G f \right)^n$; an odd number see nothing. The $(2n)!$ orderings produce each of the $(2n)!/(2^n n!)$ pairings $2^n n!$ times, so the prefactor is $i^{-2n} i^n = i^{-n}$.